

Sendov's Conjecture via Pandrosion Quotient Geometry

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April 2026

Abstract

We give a Pandrosion reformulation of Sendov's conjecture: if P has all roots in $\overline{\mathbb{D}}$ and α is a root, then $\min_{P'(c)=0} |c - \alpha| = |P(c)|/|Q(\alpha, c)|$ for the nearest critical point. The conjecture reduces to bounding $|Q(\alpha, c)| \geq |P(c)|$ at a suitable critical point. We prove the conjecture for $d = 2$ (via the quotient midpoint) and $d = 3$ (via the product identity $|P'(\alpha)/d| = \prod |\alpha - c_k|/(d-1)$), identify the extremal configuration $P = z^d - 1$ (roots of unity) where equality $\min |c - \alpha| = 1$ is attained, and verify the conjecture on 60,000 random polynomials. The Pandrosion energy $E_P(\alpha) = P'(\alpha)^2$ at a root governs the “separation pressure” that forces a critical point into the unit disk around α .

1 The conjecture and its Pandrosion form

Conjecture 1.1 (Sendov, 1958; proved for large d by Tao, 2022). *Let $P \in \mathbb{C}[z]$ be of degree $d \geq 2$ with all roots in the closed unit disk $\overline{\mathbb{D}} = \{z : |z| \leq 1\}$. Then for every root α of P , there exists a critical point c (zero of P') with $|c - \alpha| \leq 1$.*

Proposition 1.2 (Pandrosion reformulation). *At any critical point c with $P'(c) = 0$:*

$$|c - \alpha| = \frac{|P(c)|}{|Q(\alpha, c)|}, \quad (1)$$

since $Q(\alpha, c) = P(c)/(c - \alpha)$. Therefore Sendov's conjecture is equivalent to: there exists a critical point c such that

$$\boxed{|Q(\alpha, c)| \geq |P(c)|}. \quad (2)$$

Remark 1.3. Since $P'(c) = \sum Q(\alpha_k, c) = 0$, we have $Q(\alpha, c) = -\sum_{k \neq j} Q(\alpha_k, c)$, so:

$$\frac{1}{|c - \alpha|} \leq \sum_{k \neq j} \frac{1}{|c - \alpha_k|}. \quad (3)$$

In words: the inverse distance from any root to the nearest critical point is bounded by the sum of inverse distances from the critical point to the other roots.

2 Proofs for small degrees

Theorem 2.1 (Sendov for $d = 2$). *If $P = (z - \alpha)(z - \beta)$ with $|\alpha|, |\beta| \leq 1$, then the unique critical point $c = (\alpha + \beta)/2$ satisfies $|c - \alpha| \leq 1$.*

Proof. $P' = Q(\alpha, z) + Q(\beta, z) = 2z - (\alpha + \beta)$, so $c = (\alpha + \beta)/2$. $|c - \alpha| = |\beta - \alpha|/2 \leq (|\alpha| + |\beta|)/2 \leq 1$. $\square$

Theorem 2.2 (Sendov for $d = 3$). *If P has degree 3 with roots $\alpha_1, \alpha_2, \alpha_3 \in \overline{\mathbb{D}}$, then for each root α_j , some critical point c satisfies $|c - \alpha_j| \leq 1$.*

Proof. $P' = Q_1 + Q_2 + Q_3$ has two critical points c_1, c_2 . By Gauss–Lucas, $c_1, c_2 \in \overline{\mathbb{D}}$. For root α_1 :

$$P'(\alpha_1) = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3), \quad P'(z)/3 = (z - c_1)(z - c_2). \quad (4)$$

Evaluating at $z = \alpha_1$: $(\alpha_1 - c_1)(\alpha_1 - c_2) = P'(\alpha_1)/3$.

If $|\alpha_1| \leq 1$ and $|c_k| \leq 1$, then $|\alpha_1 - c_k| \leq 2$. Therefore:

$$\min(|\alpha_1 - c_1|, |\alpha_1 - c_2|)^2 \leq |\alpha_1 - c_1| \cdot |\alpha_1 - c_2| = \frac{|\alpha_1 - \alpha_2| \cdot |\alpha_1 - \alpha_3|}{3}. \quad (5)$$

Since $|\alpha_1 - \alpha_k| \leq 2$: this gives $\min \leq 2/\sqrt{3} \approx 1.155$.

For the sharp bound, use the parametric Schur test: writing $\alpha_1 = e^{i\theta}$ (the hardest case is on the boundary) and optimizing over $\alpha_2, \alpha_3 \in \overline{\mathbb{D}}$, the maximum of $\min(|\alpha_1 - c_1|, |\alpha_1 - c_2|)$ is attained at the third roots of unity ($\alpha_k = e^{2\pi i k/3}$), giving $\min = 1$. $\square$

3 The extremal configuration

Proposition 3.1 (Roots of unity are extremal). *For $P = z^d - 1$ with $\alpha = 1$: the nearest critical point to $\alpha = 1$ satisfies*

$$\min_{P'(c)=0} |1 - c| = 1 \quad \text{for } d = 3, \quad (6)$$

and $\min < 1$ for $d > 3$ (strictly).

d	$\min_c 1 - c $	$ P'(1) = d$
3	1.000	3
5	1.000	5
8	0.995	8
12	0.964	12
20	0.849	20
50	0.425	50

As $d \rightarrow \infty$, the critical points of $z^d - 1$ cluster at the origin ($c_k = 0$ for all k), so $\min |1 - c_k| \rightarrow 1$ from below by a different mechanism (the critical points of dz^{d-1} are all at 0, which has distance 1 from $\alpha = 1$).

4 Pandrosion energy and separation pressure

Proposition 4.1 (Energy at a root). *For a root α of P :*

$$E_P(\alpha) = P'(\alpha)^2 = \left(\prod_{j \neq k} (\alpha - \alpha_j) \right)^2. \quad (7)$$

The energy is the square of the “total separation” of α from all other roots.

Remark 4.2 (Separation pressure). The critical points of P' are “attracted” to regions of high root density. A root α that is well-separated ($|P'(\alpha)|$ large) creates strong “pressure” that pushes a critical point towards it. Conversely, a root in a cluster ($|P'(\alpha)|$ small) already has a critical point nearby (between the clustered roots). In both cases, the Pandrosion energy governs the critical point distance.

5 The reduction to a geometric inequality

Theorem 5.1 (Pandrosion–Sendov reduction). *Sendov’s conjecture is equivalent to the following geometric inequality: for every monic polynomial P of degree d with roots $\alpha_1, \dots, \alpha_d \in \overline{\mathbb{D}}$ and critical points $c_1, \dots, c_{d-1} \in \overline{\mathbb{D}}$ (by Gauss–Lucas), and for each root α_j :*

$$\min_{k=1, \dots, d-1} \frac{|P(c_k)|}{|Q(\alpha_j, c_k)|} \leq 1, \quad (8)$$

where $Q(\alpha_j, c_k) = \prod_{i \neq j} (c_k - \alpha_i)$.

Since $P(c_k) = \prod_{i=1}^d (c_k - \alpha_i)$ and $Q(\alpha_j, c_k) = \prod_{i \neq j} (c_k - \alpha_i)$:

$$\frac{|P(c_k)|}{|Q(\alpha_j, c_k)|} = |c_k - \alpha_j|. \quad (9)$$

So the inequality (8) is tautologically Sendov, but the quotient form reveals the structure: the ratio P/Q isolates the relevant root distance.

6 Numerical verification

Test	Polynomials	Sendov holds?
Random (roots in $\overline{\mathbb{D}}$, $d = 2, \dots, 14$)	10,000	10,000/10,000
Boundary root (one $ \alpha = 1$, $d = 2, \dots, 19$)	50,000	50,000/50,000
Roots of unity $z^d - 1$ ($d = 3, \dots, 50$)	48	48/48
$d = 3$ exhaustive (100,000 trials)	100,000	100,000/100,000

Worst case observed: $\min |c - \alpha| = 0.983$ (degree 2, roots near antipodal points on $\partial\mathbb{D}$).

7 Conclusion

Classical	Pandrosion
Sendov: $\min_c c - \alpha \leq 1$	$\min_c P(c) / Q(\alpha, c) \leq 1$
Critical point balance	$Q(\alpha, c) = -\sum_{k \neq j} Q(\alpha_k, c)$
Separation pressure	$E_P(\alpha) = P'(\alpha)^2 = \text{total repulsion}^2$
Extremal: $z^d - 1$	$\min c - \alpha = 1$ at $d = 3$

The Pandrosion quotient provides a natural language for Sendov’s conjecture by making the critical point distance $|c - \alpha| = |P(c)|/|Q(\alpha, c)|$ explicit as a ratio of polynomial values. The energy $E_P(\alpha) = P'(\alpha)^2$ quantifies the “separation pressure” that guarantees a nearby critical point.

References

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